

In this assignment, we focus on the **Pre-compile/Configuration** phase of the AUTOSAR (AUTomotive Open System ARchitecture) methodology. Here a system integrator must ensure that the software components (SWCs) and Basic Software (BSW) modules are correctly configured before the build process begins.

1. Configuration Breakdown

To complete this assignment, you need to follow a structured software validation pipeline:

Parse AUTOSAR XML (ARXML)

ARXML files are complex schemas that describe everything from hardware pin mapping to software runnables.

- **Approach:** Use an XML parser (like Python's `ElementTree` or `lxml`).
- **Key Tags:** Look for `ECU-INSTANCE`, `MODULE-CONFIGURATION`, and `PARAMETER-VALUES`.
- **Goal:** Extract the specific configuration values into a readable data structure (like a Dictionary or JSON).

Validate Required Parameters

This is the core logic. You must check the parsed data against specific "Golden Rules." Examples include:

- **Range Checks:** Is the `BaudRate` for the CAN controller between 125kbps and 1Mbps?
- **Existence Checks:** Does every `SWC` have an assigned `ECU-INSTANCE`?
- **Consistency Checks:** If `Sleep Mode` is enabled, is the `Wakeup Source` also defined?

Generate Validation Report

A professional report should categorize findings to help developers debug the configuration.

- **PASS:** Parameter meets all constraints.
- **WARNING:** Parameter is valid but suboptimal (e.g., high memory usage).
- **FAIL:** Invalid configuration that will cause a system crash or build error.

Mapping the code to the three required layers of automotive software regulation:

Layer	Module Function	Responsibility
Data Collection	<code>ws.loadXML()</code>	The Perception phase where the workspace ingests the digital environment.
Data Processing	<code>ws.find()</code>	The Decision phase where specific objects are identified and filtered.
Predictive Analysis	<code>if min <= val <= max</code>	The Action phase where the system predicts if the config is safe for the physical ECU.

Flowchart:

